Short-Term Fog Forecasting using Meteorological Observations at Airports in North India

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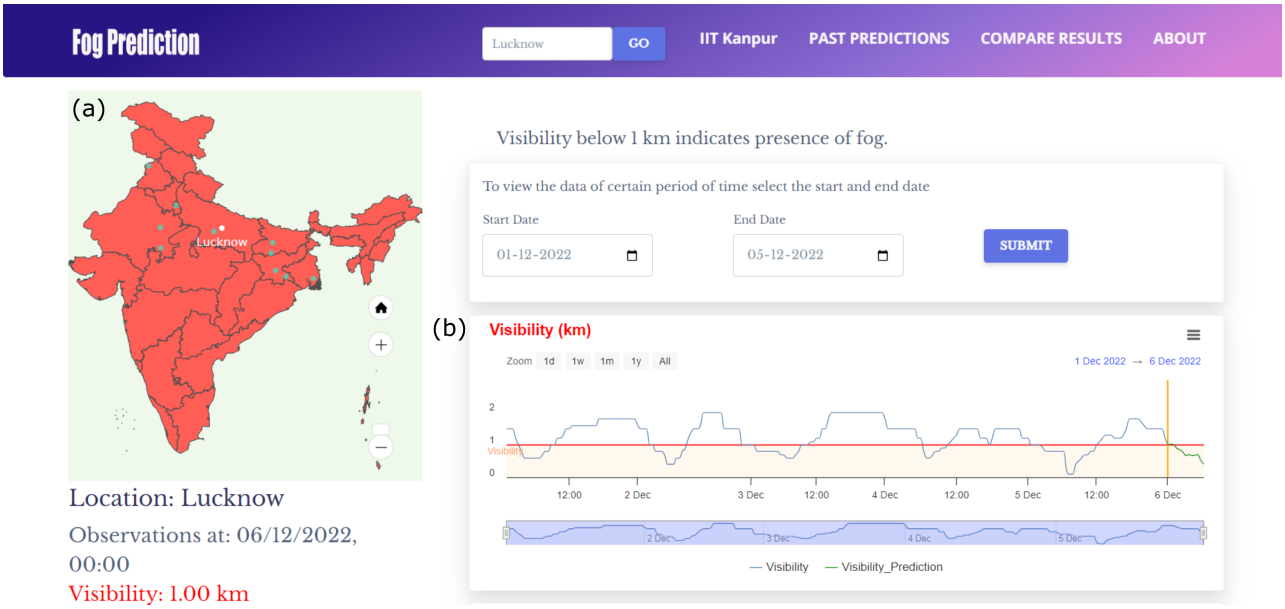


Figure 1: Fog prediction website snapshot (for a past date) showing (a) cities where we forecast fog, (b) visibility (fog) prediction for next 3 hours (blue: past visibility, green: visibility forecast, red: 1-km visibility line separating fog and no-fog, yellow: current time)

ABSTRACT

Fog is a major phenomenon in North India during the winter months of November to February. It lowers visibility in these regions which adversely affects surface and air transportation, agriculture, and other day-to-day activities. Flights get delayed, canceled or diverted due to reduced visibility conditions. Therefore, accurate forecasting of fog (which is measured as visibility in terms of distance) at the airports up to 6 hours is important. We develop data science models that use historical meteorological observations (1974-2021) from METAR reports of ground stations at different airports of North India for short-term fog prediction. Models are designed for both binary classification ("fog" versus "no-fog") as well as multi-class classification (that categorizes fog into 4 classes). In addition, we also design regression models that predict visibility. Results show that the best visibility regression model has a root-mean-squared-error (RMSE) of 0.20 km for 3-hour lead time prediction for Lucknow airport. Corresponding classification accuracies are 0.89 and 0.79 for the 2-class and 5-class problems respectively. Similar trends were observed for other airports and lead times. A web-based dissemination system has been deployed at https://fog.iitk.ac.in.

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CCS CONCEPTS

• **Applied computing** → **Forecasting**; • **Mathematics of computing** → *Time series analysis*; • **Computing methodologies** → *Machine learning*.

KEYWORDS

Fog Forecast, Visibility, Data Science, Time-series, Nowcasting

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1 INTRODUCTION AND MOTIVATION

Fog is a suspension of tiny, usually microscopic water droplets in the air that reduces visibility at the earth's surface to below 1 km [19]. It causes disturbances in air, rail, and road transport services, and elevates risk of accidents. In winter, fog frequently occurs in North India causing significant economic losses (for instance, it cost the Delhi airport a loss of over 3.9 billion USD between 2011 and 2016 [23]). Other sectors such as agriculture are also impacted by fog. Fog's detrimental impacts can be partly mitigated by its accurate forecasting [4].

Fog is measured in terms of *visibility*. The visibility sensors measure *meteorological optical range* defined as a distance required to reduce the luminous flux in a collimated beam from an incandescent lamp to 5% of its original value. *Fog forecasting* is typically carried out by numerical weather prediction (NWP) models or statistical models. The NWP models forecast the weather by numerically solving conservation equations for fluid flow. Since NWP models do not estimate visibility, fog is forecasted by placing deterministic thresholds on the predictions of basic weather variables like temperature and relative humidity. Computational NWP models cannot respond well to irregular data inputs as they rely on deterministic formulae. This poses a severe limitation on the accuracy of fog forecasts. For example, the probability of detection (PoD) varies from 29-48% for North Indian cities using the NWP model [28]. The statistical models (and data science or AI/ML methods) aim to learn the patterns of the meteorological variables for fog forecasting and have achieved relatively better results [9].

A real-time fog and visibility forecast system is not yet developed for North India, which motivated this work. Our objectives and contributions in this paper are:

1. To develop fog (i.e., visibility) prediction algorithms;
2. To assess their performance vis-a-vis existing/baseline models;
3. To implement the algorithms for real-time prediction and dissemination of the prediction results on a website.

2 RELATED WORK

Fog forecast studies over India are reviewed. Next, studies that have used machine learning models or their ensembles, time-series models, and outputs of NWP models are reviewed. Finally,

Fog forecast for Bangalore airport was developed using correlation coefficients between frequency of fog and cloud cover, relative humidity and wind speed [25]. A model based on statistical multiple discriminatory analysis was used for predicting fog over Delhi Roy Bhomick et al. [26]. For Kolkata airport, an ANN network was proposed to forecast fog at 3-hour lead time [17]. The predictors for the model were identified using a decision-tree algorithm. Srivastava et al. [29] developed an ARIMA model for Ghaziabad city to forecast number of fog days in a winter season (October to February) for five years in advance, i.e., 2015-2020.

Colabone et al. [11] forecasted presence/absence of fog at airport in Pirassununga, São Paulo State, Brazil using multilayer perceptrons. Deng et al. [13] used long short-term memory (LSTM) networks for forecasting visibility at Beijing airport. The model performed well for short-term prediction (3-hour), but the performance deteriorated for long-term prediction (greater than 6-hour). Zhu et al. [33] used hourly observations from 2007-2016 of Urumqi International Airport, China to predict visibility ranges. Dietz et al. [16] proposed ordered logistic regression to forecast probabilities for each LVP (low visibility procedures) states at the Vienna International Airport.They analyzed nowcasts, medium-range forecasts using boosted trees to predict LVP states with a lead time of 12 days. Bartok et al. [4] improved visibility prediction application in Dubai using decision tree induction for 6 hour lead time. The model delivered better results compared to preexisting NWP models. Dewi et al. [14] used stacked ensemble to predict fog and non-fog events at 3-hour lead time at Wamena Airport, Indonesia. The ensemble consisted of distributed random forest, deep learning, gradient boosting machine, generalized linear model and extreme randomized tree.

An Auto-regressive Integrated Moving Average (ARIMA) model was developed to forecast visibility at Hang Nadim Airport, Batam, Indonesia [27]. AIC and BIC were used for selecting the best ARIMA model. An Auto Regressive Recurrent Neural Network (ARRNN) was proposed to forecast visibility at Dulles International Airport, Virginia [22]. The ARNN model was shown to peform better than LSTM and vanila RNN is predicting visibility. Cornejo-Bueno et al. [12] developed support vector regression, neural network, and Gaussian process based models for predicting of low-visibility events at Valladolid airport, Spain. Guijo-Rubio et al. [18] developed a hybrid auto-regressive prediction model that combines fixed-size and dynamic windows, to model the dynamics of the visibility time series and classify fog three ordinal categories, 'fog', 'mist' and 'clear' for the same airport.

Bari [3] analysed sensitivity of various data-mining techniques, especially gradient-boosting regression and deep-learning approaches, applied to NWP model outputs to improve visibility prediction over multiple stations in Morocco. K-nearest neighbors clustering (KNN), gradient boosting, random forest and support vector classifier algorithm were developed using *reforecasts* from an NWP model and tested over four airports in the United States of America [21].

3 STUDY AREA AND DATASETS

The study area for our work is Indo-Gangetic Plains (IGP) in North India. Ten cities spread over IGP are studied (Fig 1 presents their locations). They are Lucknow, Amritsar, Delhi, Jaipur, Patna, Gaya, Jamshedpur, Kolkata, Ranchi, and Kota. In addition, we have also set up a weather monitoring station at IIT Kanpur. The elevations of

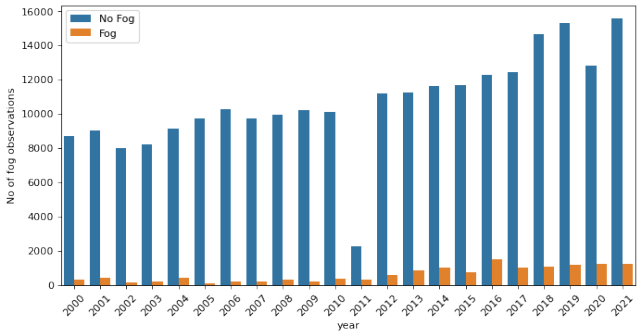


Figure 2: Yearly variation of observations at Lucknow

Test	Year	# Fog	# No Fog	Total
Training	2000-2015	9,746	69,145	78,891
Validation	2016-2018	4,005	12,383	16,388
Testing	2019-2023	5,683	16,379	22,062
Total	2000-2023	19,434	97,907	**1,17,341**

Table 1: Data statistics for Lucknow after preprocessing

Binary	Multi-class	Visibility (m)
No Fog	No Fog	≥1001
Fog	Shallow Fog	501-1000
	Moderate Fog	201-500
	Dense Fog	51-200
	Very Dense Fog	≤50

Table 2: Fog classes and associated visibility ranges

the cities (above mean sea level) range from 5 m (Kolkata) to 651 m (Ranchi).

The visibility observations are obtained from METAR records (except at IIT Kanpur where we measured visibility using a forward scatter sensor). These are periodic weather reports issued by the airports and are available near-real-time on the Iowa Environmental Mesonet (IEM) archive (https://mesonet.agron.iastate.edu/request/download.phtml). The METAR observations contain half-hourly measurements of visibility, dew point temperature, rainfall, air temperature, relative humidity, wind speed, and air pressure at the ground surface. These METAR observations are chosen for study as they are measured *in-situ* and are, hence, more reliable and trustworthy. The observations from 1972 to 2021 for November to February (i.e., winter months) are used.

Fog is measured using visibility. However, there are certain non-fog situations that can also create low visibility, e.g., dust storms [15], heavy rainfall, high air pollution [31], etc. In such cases, along with visibility, corresponding meteorological parameters such as relative humidity, etc. becomes an important variable.

The collected data is not recorded consistently over the years, as can be seen in Fig. 2. For the years 1972-1995, the frequency of data is quite low and increases evenly from 1996 onwards up to 2011 when the frequency becomes less again. Maximum observations are recorded for winter of 2019.

Therefore, the entire dataset is subdivided into two datasets, DS1 and DS2 based on time-period. DS1 consists of the entire data, and has a training period from 1972-2015 and testing period from 2016-2021. DS2 consists of data from the year 2000 onwards has a training period from 2000-2018 and testing period from 2019-2023. A validation set consisting of years 2016-2018 was carved out of the training set. DS1 has a larger proportion of missing values, in addition to a lesser frequency of time sampling. Hence, for more reliable models, by default, we use and present results on DS2.

4 METHODOLOGY

4.1 Problem Statement

Formally, for each time instance t_i, we have a tuple of values $\mathcal{V}_i = \langle t_i, v_i, \{m_i^{(1)}, m_i^{(2)}, \dots\}\rangle$ where $v_i \in \mathcal{R}^+$ denotes the visibility value, and $m_i^{(1)}$ etc. denote the values of different meteorological (and other) variables. The entire time-series dataset is represented as an ordered set $\mathcal{D}$ of such tuples $\mathcal{V}_i$ up to the current time t_0.

The problem is to predict the visibility at a future lead time point t_l, given $\mathcal{D}$ till time t_0. The *regression* problem is: given a lead-time of l, predict v_l. Corresponding to each visibility value v_i, there is an associated class c_i, according to Table 2. We have followed the standard based on International Civil Aviation Organization (ICAO). For the binary classification problem, c_i can be of 2 possible values (all the different fog classes are clubbed into the "fog" class), while for the multi-class problem, it has 5 possibilities. The *classification* problem is: given a lead-time of l, predict c_l.

4.2 Evaluation Metrics

We use the following metrics to evaluate the regression models:

(1) MAE (Mean Absolute Error): MAE calculates absolute difference between actual and predicted values.
(2) RMSE (Root Mean Squared Error): It is the square root of squared difference between actual and predicted values.

The metrics for the classification problem are:

(1) Accuracy: A prediction is accurate if its class is equal to the actual. We measure both the accuracy over all test points as well as a classwise average.
(2) (Macro) F1-score: Macro F-score is the average F1-score across all the classes.

4.3 Pre-processing

To ensure uniformity, all the *in-situ* observations are initially converted into SI units. Since we work with real data, these observations have missing and erroneous values. Thus, it is important to preprocess the dataset.

4.3.1 Normalization and Resampling. First, erroneous observations are identified which correspond to values that are physically impossible, e.g., relative humidity >100%, visibility <0 km, dew point temperature above air temperature, etc., and are removed. Appealing to domain knowledge, we next identify those observations

that may be physically possible but extremely rare and anomalous in view of the usual IGP weather. These include sudden changes within 30 minutes in temperature by 50°C or wind speed by 100 kmph, etc. These observations are replaced with null tokens, signifying missing values. After that, observations that are duplicated in terms of time-stamp are removed. Finally, the dataset is scaled using min-max normalization. To make the data uniformly sampled, the time series is resampled to 0.5 hr.

4.3.2 Interpolation. Most data science models assume that there are no missing values and, hence, we devise ways to impute the missing values. We try the following methods of imputation: (1) linear interpolation, and (2) spline interpolation. We evaluate the missing value imputation methods by masking some of the real values present, and running both the algorithms. Linear interpolation method performed the best in terms of squared error (details in Sec. 5) and, hence, we used that.

If, however, the range of values that are missing is more than a window size of 3 hours, we do not attempt to impute them, since then there are too many values missing. Instead, we delete that whole time-chunk of values from the dataset.

4.3.3 Stationarity. *Stationarity* is a property of a time series that refers to it being time independent. More precisely, the mean, variance and auto-correlation of a stationary series is constant over time. Auto-correlation is the correlation of the series with its past values, i.e., how values at time $t + k$ are related to values at time t. Partial auto-correlation function (PACF) conveys pure correlation between a value and the corresponding target lag value, excluding the effect of intermediate lags.

Stationarity of a time series can be understood by looking at its plot. However, to quantitatively measure stationarity, some statistical tests known as Unit Root Tests are employed. We tried three such tests: (1) Augmented Dickey Fuller (ADF) test, (2) Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test, and (3) Philips Perron (PP) test. All the tests showed that the original data is non-stationary.

A non-stationary series can be made stationary by differencing, which involves subtracting the previous value $t - 1$ from the current value t. If first-order differencing does not make the series stationary, then the resulting differenced series can be differenced again, and so on, till the series becomes stationary. In practice, up to second order differencing is sufficient for most real datasets. Stationarizing, thus, aims to remove any correlation present in the data.

We applied second-order differencing before applying time-series models such as ARIMA.

4.3.4 Seasonality. As shown in Fig. 3, North India experiences fog events almost exclusively during the winter months of November, December, January, and February. So, only these months (Nov-Feb) are considered for this study. To analyse the seasonality present in time-series, additive decomposition was performed which breaks down the series into 4 components – seasonal, trend, cyclical and noise. This shows that there is no seasonality present in the data for the winter months. The visibility values begin falling from November and reach their lowest levels in January after which they rise in February as fog events dissipate. The pattern repeats itself approximately every year. The trend component of the additive decomposition gave similar insights.

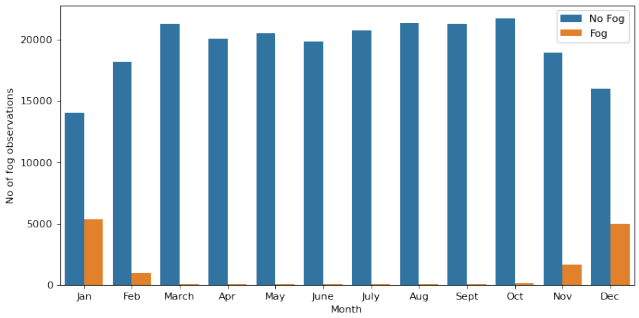


Figure 3: Monthly variation of observations at Lucknow

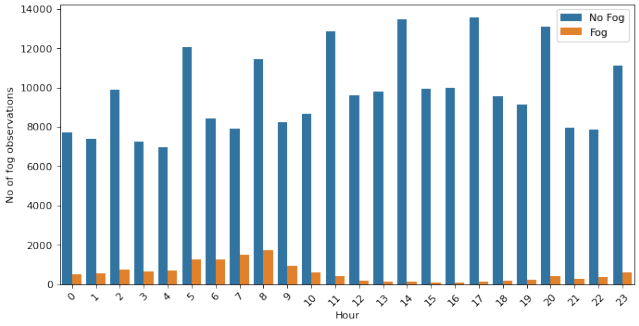


Figure 4: Hourly variation of observations at Lucknow

4.4 Handling Class Imbalance

We observed heavy class imbalance since a lot of non-fog observations were present as compared to fog events (refer to Table 1 for data statistics). Therefore, we experimented with three balancing strategies:

(1) Oversampling minority classes: SMOTE (SMT) [10], or Borderline SMOTE (BSM) [20]
(2) Undersampling the majority class: Condensed Nearest Neighbours (CoNN) [2], or Neighbourhood Cleaning Rule (NCR) [1], or Tomek Links (TL) [32], or Random Under Sampler (RUS)
(3) Hybrid method: Undersampling the majority class and oversampling minority classes

4.5 Feature Importance

To assess the importance of features, we used Shapley values. For an individual instance, the Shapley value of a feature indicates how important it is in predicting the class (or regression value) of the instance.

4.6 Feature Extraction

While it is likely to have fog during the winter months, not all hours of the day have a uniform probability of having fog. Fig. 4 shows a cyclical behavior of fog throughout the day. We observe peak fog hours in early morning, after which it dissipates during daytime. Fog again starts accumulating from late evening. To accommodate this behavior of fog, we augment the values by *sine* and *cosine* functions of the minute value corresponding to the time of the day and include

them as features:

$$\text{cos-time} = \cos(2\pi \cdot \text{minute-time}/1440)$$
$$\text{sin-time} = \sin(2\pi \cdot \text{minute-time}/1440)$$

This transformation performs cyclical encoding of the time of day.

4.7 Baseline Persistence Model

We implemented a simple baseline, the *persistence* model, where we assume that the current visibility condition will persist for a lead time of l. In other words, the model predicts $v_l = v_0$ (and similarly, $c_l = c_0$). This is a strong baseline, especially for shorter lead times.

4.8 Classification and Regression Models

We tried a number of models for classification and regression. These include standard machine learning models [7], ensemble methods [7], time series methods [8], [5], and sequence models [6].

Machine learning include Logistic Regression (LR), Decision Tree (DT), Multi-layer Perceptron (MLP), Neural Network (NN), Support Vector Machine (SVM), Naive Bayes Classifier (NB), and K Nearest Neighbors (KN).

Some of the ensemble methods we tried are Gradient Boosting Classifier (GBC), Random Forest (RF), XG Boost (XGB), and AdaBoost (ADC).

Auto-Regressive Integrated Moving Average (ARIMA) and Hidden Markov Model (HMM) time series models were also tested.

Sequence models represent the class of recurrent neural network (RNN) models that model sequences; we tried Long Short Term Memory (LSTM) and Gated Recurrent Unit (GRU). Stacked and bidirected variants of these models were also tested.

The hyper-parameters for the machine learning, ensemble and sequence models were tuned using a grid search method using MAE as the monitoring metric over the validation set. ARIMA models used the AIC (Akaike Information Criterion), which is a measure of prediction error [30] as the metric to optimize. HMM have the best results on the validation set for 2 hidden states.

5 RESULTS AND DISCUSSION

5.1 Preprocessing

During preprocessing, we first de-duplicated the data. The duplicates were found in 3 cities only and were less than 10 in each city. Next, erroneous observations were identified. Such observations amounted to less than 0.01% of the total observations.

The percentage of missing values was around 2%. As explained earlier, we tested both linear interpolation and spline interpolation on the true values in the validation set. Linear interpolation method performed better and were, hence, used for the entire data. Missing value imputation and resampling increased the number of data points corresponding to Lucknow from 83,471 to 1,17,341, i.e., an increase of more than 40%.

5.2 Class Imbalance

We applied all the three methods (undersampling, oversampling and hybrid) as discussed in Sec. 4.4 to handle class imbalance. The best results for undersampling were obtained for the NCR strategy [1] using only the majority class, with 3 nearest neighbors and a threshold of 0.5. Simple SMOTE [10] was the best oversampling strategy.

Running NCR on the majority no-fog class reduced the number of instances from 69,145 to 67,016. SMOTE for binary classification problem increased the number of instances for the fog class from 9746 to 50% of the majority class, i.e., 31,780. A slightly different strategy was used for the multi-class problem, where all the fog classes were increased to the most frequent class (which was shallow fog having 5,089 observations). Thus, after oversampling, all the 4 fog classes had 5,089 instances each.

Model Name	Accuracy	Macro F1	F1-score of Fog
Persistence	*0.85*	*0.78*	*0.71*
MLP NCR	0.88	**0.85**	**0.77**
LR SMT	0.86	0.84	**0.77**
SVM SMT	0.87	0.84	**0.77**
LR SMT TL	0.86	0.84	**0.77**
SVM SMT TL	0.87	0.84	**0.77**
LR NCR	0.88	0.84	**0.77**
MLP	**0.89**	0.84	0.76
GBC	0.88	0.83	0.74
ADC	0.88	0.83	0.75

Table 3: Binary classification results for 3 hr lead time: Lucknow

		Predicted		
		Fog	**No Fog**	**Total**
Actual	**Fog**	4901	782	5683
	No Fog	2403	13976	16379
	Total	7304	14758	22062

Table 4: Confusion matrix (3-hr lead time, Lucknow, MLP-NCR)

5.3 Binary Classification Results

Table 3 shows the performance of different models used for binary classification over Lucknow for a lead time of 3 hours. The best model in terms of macro-F1 score was multilayer perceptron with undersampling using neighborhood cleaning rule (MLP-NCR).

As stated earlier, a grid search was performed on the validation set to find the best hyperparameters. They are: (a) No of hidden layers: 1, (b) Number of neurons in the hidden layer: 100, (c) Optimizer: Adam, (d) Activation function: Sigmoid, (e) Maximum number of iterations: 200, (f) Early stopping: true. Table 4 shows the corresponding confusion matrix for the binary class problem.

We also applied this best model for other lead times. The results are shown in Fig. 5. As expected, both accuracy and F-score fall with increasing lead times.

The results for other cities are similar to that of Lucknow, as shown in Fig. 6.

5.4 Multiclass Classification Results

As shown in Table 5, the best performing model in terms of macro F1-score for the 5-class classification problem for a lead time of

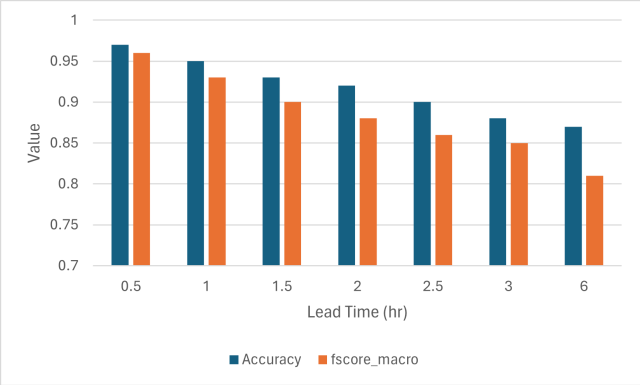


Figure 5: Results for different lead times for Lucknow

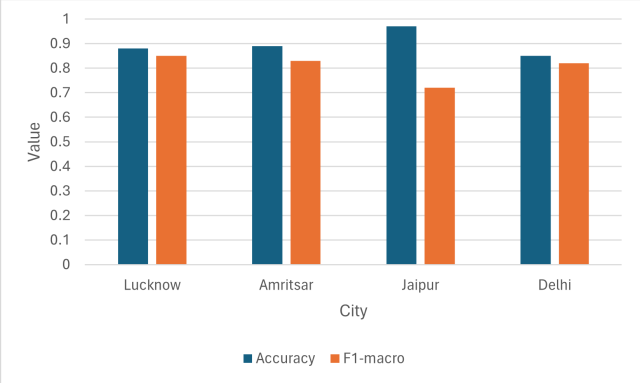


Figure 6: Results for 3 hr lead time for multiple cities

Model Name	Accuracy	Macro-F1	Macro-F1 of Fog
Persistence	*0.77*	*0.45*	*0.30*
GBC-NCR-SMT	0.79	**0.49**	**0.38**
GBC	0.79	0.45	0.34
GBC-SMT	**0.80**	0.46	0.34
GBC-OVR	**0.80**	0.47	0.35
LR	0.68	0.38	0.26
LR-OVR	0.68	0.38	0.26
RF-OVR	0.67	0.45	0.36
RF	0.62	0.42	0.32
DT-OVR	0.62	0.38	0.27
DT	0.49	0.33	0.23

Table 5: Multi-class results for 3 hr lead time: Lucknow

3 hours for Lucknow is Gradient Boosting after applying a hybrid class balancing strategy using both NCR and SMOTE. We call this model GBC-NCR-SMT. The best hyperparameters (found using grid search) are learning rate: 0.05, loss: deviance, maximum depth: 3, number of trees: 120, and subsampling rate: 0.5.

The table also shows the average F1-score of the 4 fog classes.

		Predicted					
		VD	**D**	**M**	**S**	**NF**	**Total**
Actual	**VD**	257	99	108	59	18	541
	D	109	129	156	100	64	558
	M	34	117	353	198	197	899
	S	22	65	627	1268	1703	3685
	NF	4	11	109	732	15523	16379
	Total	426	421	1353	2357	17505	22062

Table 6: Multi Class Confusion Matrix (3-hr lead time, Lucknow); VD = Very Dense Fog, D = Dense Fog, M = Moderate Fog, S = Shallow Fog, NF = No-Fog

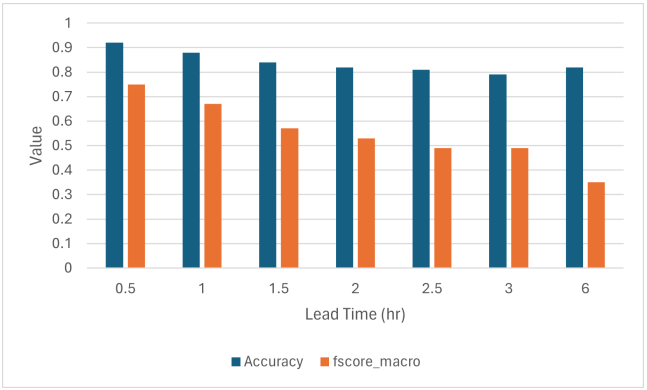


Figure 7: Multi-class results for all lead times: Lucknow

The corresponding confusion matrix is shown in Table 6. Most of the errors happen between the neighboring fog classes, which is expected.

Fig. 7 shows results of the best classification model for all lead times over Lucknow. It shows that, as expected, it becomes harder to classify with increasing lead times.

The results for other cities are similar and are, hence, omitted.

5.5 Regression Results

Table 7 shows summary of the experimental results for various models for regression for a lead time of 3 hours on Lucknow data.

The lead times considered in the current analysis are (in hours) - [0.5, 1, 1.5, 2, 2.5, 3, 6]. We note that data-driven models achieve quite better performance compared to the persistence model for fog forecasting. The best performing model – Univariate Vanilla LSTM has number of neurons as 128, activation function ReLU, Optimizer Adam, convergence metric was set as MSE, batch size as 64 with validation on a held-out set.The model was trained for 12 epochs with stopping criteria being no change in validation loss for 3 iterations.

We also showed that the predictions are better when the original value is less than 1 km, i.e., within the fog range. LSTM models can predict upto 100 m of the value.

Regression result for 0.5 hr to 3 hr lead time for the best performing model are in Fig. 8. We note that error metrics increase as we move to smaller lead times since the data sequence length required

Model Name	RMSE (km)	Fog RMSE	MAE (km)
Persistence	*0.68*	0.60	*0.48*
Univariate Vanilla LSTM	**0.20**	**0.14**	0.11
Stacked LSTM	0.20	0.17	**0.10**
Bidirectional LSTM	0.21	0.16	0.13
Stacked GRU	0.21	0.15	0.12
Univariate Vanilla GRU	0.22	0.15	0.13
Bidirectional GRU	0.22	0.16	0.12
LightGBM Custom Scoring	0.48	0.31	0.36
PloyLR	0.55	0.34	0.27
LightGBM	0.48	0.37	0.29
MLP	0.49	0.42	0.33
LR	0.60	0.42	0.34
GBR	0.44	0.44	0.34
RF	0.57	0.47	0.38
DT	0.58	0.48	0.39
HMM	0.50	1.05	0.98
ARIMA	1.43	2.12	1.22

Table 7: Regression results for 3-hr lead time at Lucknow

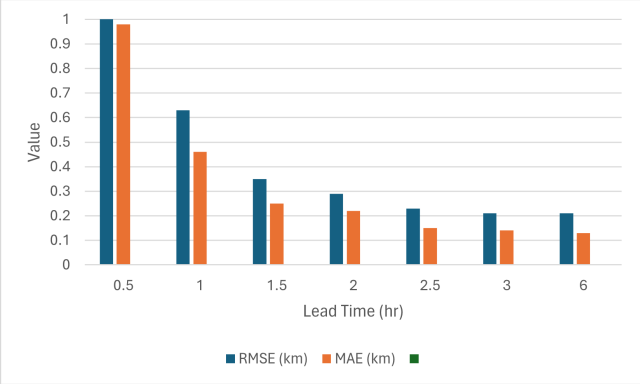


Figure 8: Result for all lead times: Lucknow

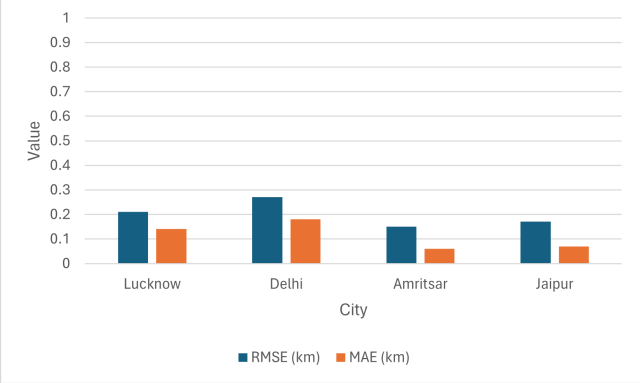


Figure 9: Regression results for 3-hr lead time for multiple cities

for forecasting decreases corresponding to the lead time which in turn provides less information about the series to the model. Fig. 9 shows a similar experimental model's performance for 3 hours lead time over other cities in study area.

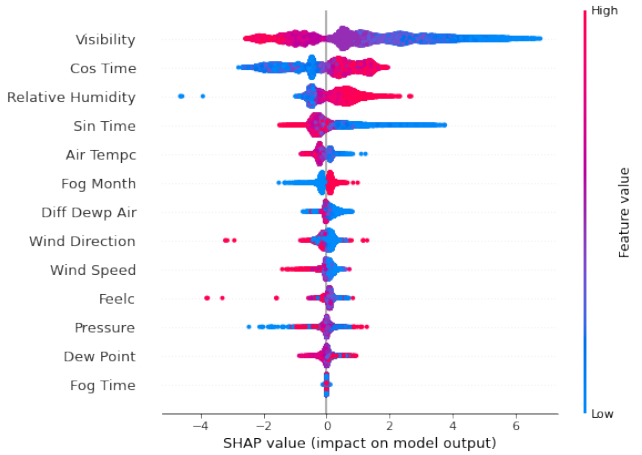


Figure 10: Violin plot showing average Shapley values for the fog class in binary classification

Our experiments showed that the lowest RMSE achieved for Lucknow is 0.21 km for three-hour lead time using a Univariate LSTM. Similarly, for Delhi, the lowest RMSE is 0.27 km for a three-hour lead time using a single-layered GRU/LSTM.

Deep learning (DL) based models outperform traditional statistical models since DL models try to learn features in an incremental manner. Statistical models are online models, and so they have higher training time as well as inference time, while DL-based models are fast-performing ones. They also take into account the sequential nature of data.

Multivariate variants with the same architecture of a model show lower performance compared to their univariate counterparts as RMSE for 3 hour lead time for Univariate LSTM is 0.21 km, whereas the RMSE for Multivariate LSTM is 0.79 km. Similarly, for Stacked LSTM in multivariate case, RMSE is 0.62 km and in the univariate case is 0.20 km. Bidirectional LSTM for multivariate fog prediction experiments give RMSE of 0.83 km whereas for univariate experiments, we obtain RMSE of 0.21 km.

5.6 Feature Selection and Feature Extraction

SHAP is a mathematical method to explain the predictions of machine learning models. The goal of SHAP is to explain the prediction of an instance x by computing the contribution of each feature to the prediction. We used the TreeSHAP [24] estimation approach.

Fig. 10 shows average feature importance with feature effects for all the test observations. The features are ordered according to their importance. Each point on the summary plot is a Shapley value for a feature and an instance. The most important feature is visibility at the current time point. This also explains why the persistence model is a strong baseline. The time of the day proves to be an important factor in prediction as well, followed by relative humidity. A high relative humidity increases the chance of an observation being fog.

5.7 Other Results

5.7.1 Using PM2.5 Data. We trained an XGBoost model to experiment with PM2.5. Using PM2.5 along with other weather and

Model Name	Accuracy	Macro F1
Gradient Boost	**0.86**	**0.75**
XGB	0.86	0.74
AdaBoost	0.86	0.74
CatBoost	0.86	0.74
NN	0.86	0.72
RF	0.85	0.73
LightGBM	0.74	0.63
DT	0.81	0.69
LR	0.84	0.62

Table 8: Fog Onset and Dissipation

		Predicted				
		0	1	2	3	**Total**
Actual	0	13784	924	0	0	14708
	1	714	949	0	0	1663
	2	17	0	1535	117	1669
	3	80	0	1054	2888	4022
	Total	14595	1873	2589	3005	22062

Table 9: Confusion matrix for fog onset and dissipation

time-based attributes, we obtained RMSE of 0.58 km and MAE of 0.43 km. Without using PM2.5 as a parameter, we obtained RMSE of 0.60 km and MAE of 0.44. PM2.5 data was available from 2015-2020, so the training split was upto 2017. Test data was considered from 2018-2020. We observed that including it as a parameter did not improve the regression metrics to a high extent. This can be attributed to the low availability of data.

5.7.2 Fog Onset and Dissipation Prediction. For many applications such as flight planning, it is important to correctly predict the time when fog will onset or it will dissipate. Hence, we tried a multi-class classification problem to see if our models could correctly predict these. The target variable is created based on the following conditions:

- *Class 0*: (remains no-fog) $vis(t) >= 1$ AND $vis(t+3) >= 1$
- *Class 1*: (fog onset) $vis(t) >= 1$ AND $vis(t+3) < 1$
- *Class 2*: (fog dissipation) $vis(t) < 1$ AND $vis(t+3) >= 1$
- *Class 3*: (remains fog) $vis(t) < 1$ AND $vis(t+3) < 1$

Table 8 shows the results. The best model, Gradient Boost, could achieve a macro-F1 score of 0.75.

The corresponding confusion matrix, shown in Table 9 shows that accurately predicting the time of fog onset is a difficult problem.

5.8 Dissemination of Results

The real-time fog (and visibility) prediction results are displayed on the Fog Prediction IITK website (fog.iitk.ac.in). The homepage of the website is shown in Fig. 1. The website's features also include a comparison of different model performances, the display of real-time weather data, and past data for the chosen city from the available ones. The code will be available on request using Contact-us section.

6 CONCLUSIONS AND FUTURE WORK

Short-term (up to lead time of 6-hour) fog forecast models are developed for ten airports in North India and the results are disseminated through the website https://fog.iitk.ac.in.

The models used ground-based meteorological observations as predictors. Binary ("fog" versus "no-fog") and multi-class classification (that categorizes fog into 4 classes) models as well as regression (that forecast visibility) models were developed. The models showed reasonable performance (e.g., classification accuracy of about 0.79 and RMSE of less than 0.20 km for 3-hour lead time prediction at Lucknow airport) compared to the baseline (persistence) model and can be potentially used for improving flight operations during winter months in North India. The best results for binary classification, multi-class classification and regression were obtained from multi-layer perceptron, GBC and single-layer LSTM, respectively. Nevertheless, the developed models have limitations and we are presently exploring avenues to overcome them.

The stationarity tests showed that visibility measurements are non-stationary. However, ML models used in the study assume data to be stationary suggesting that model parameters learned from historical data may not be optimum for forecasts. Hence, methods should be explored for dynamic adjustments of model parameters. Further, visibility can be reduced because of non-fog reasons, like heavy rainfall, dust or air pollution. The proposed data science models are not capable of distinguishing these reasons and therefore may provide erroneous results. Moreover, only deterministic forecast models are presented in this paper. Since uncertainty estimates are required for informed decision-making, we are developing probabilistic forecast models.

We have developed forecast models for real-time deployment. Hence, only those predictors which were available in near-real time were selected. Nevertheless, there are other potential predictors (e.g., soil moisture, air quality parameters, synoptic scale systems like western disturbances) that could not be used in this study because they are not available in near-real-time. In future, we would like to include them in the forecast models.

The proposed models can forecast fog only at locations where *in-situ* visibility observations are available. The use of satellites and NWP models can enable fog prediction at other locations also. Further, the visibility sensors are expensive, which poses a limitation on the number of installations of these sensors. Low-cost sensor development and citizen science can also improve the coverage of fog monitoring and forecasting. In many situations, fog forecasts beyond 6-hour lead time are required. We are currently investigating methods to increase the lead time of fog forecasts for up to five days.

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